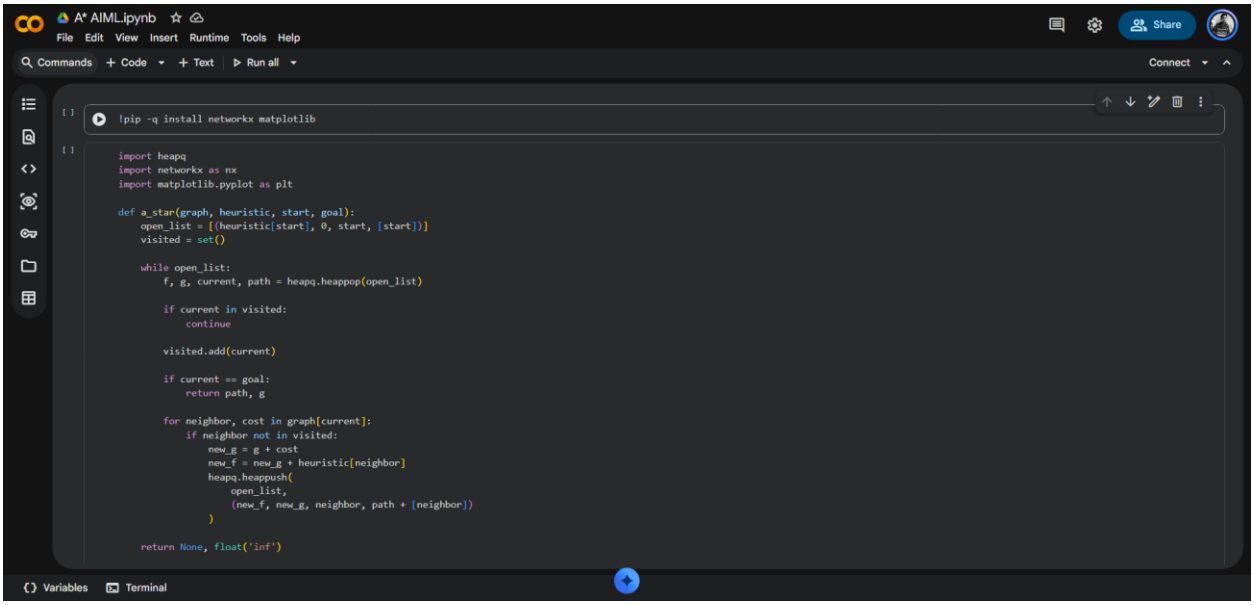
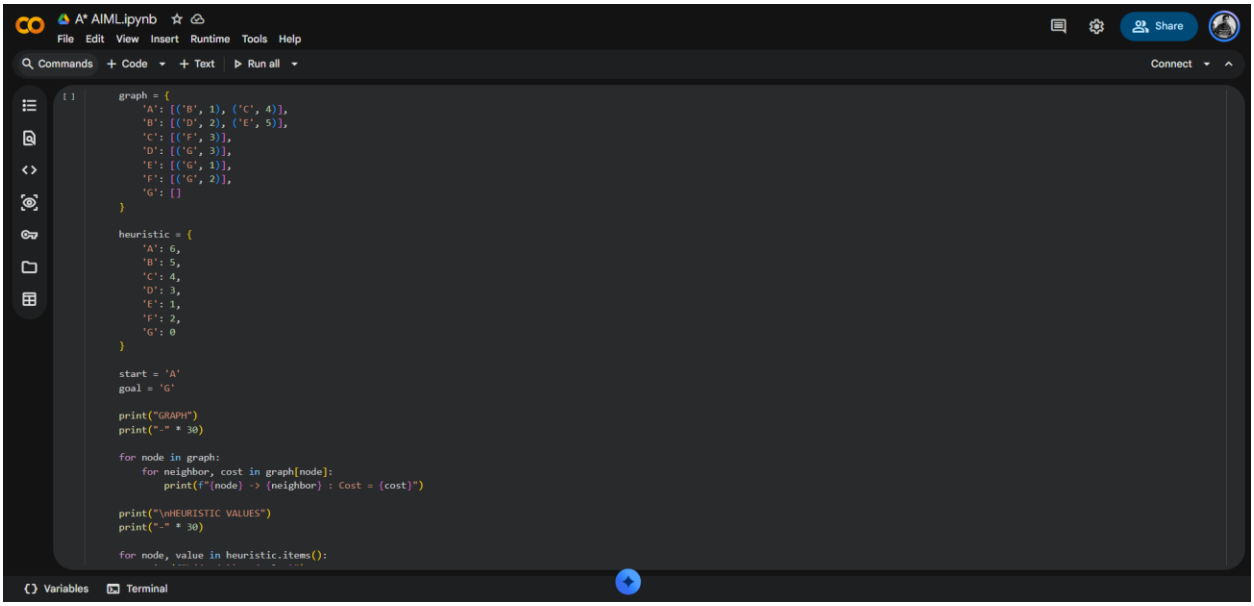
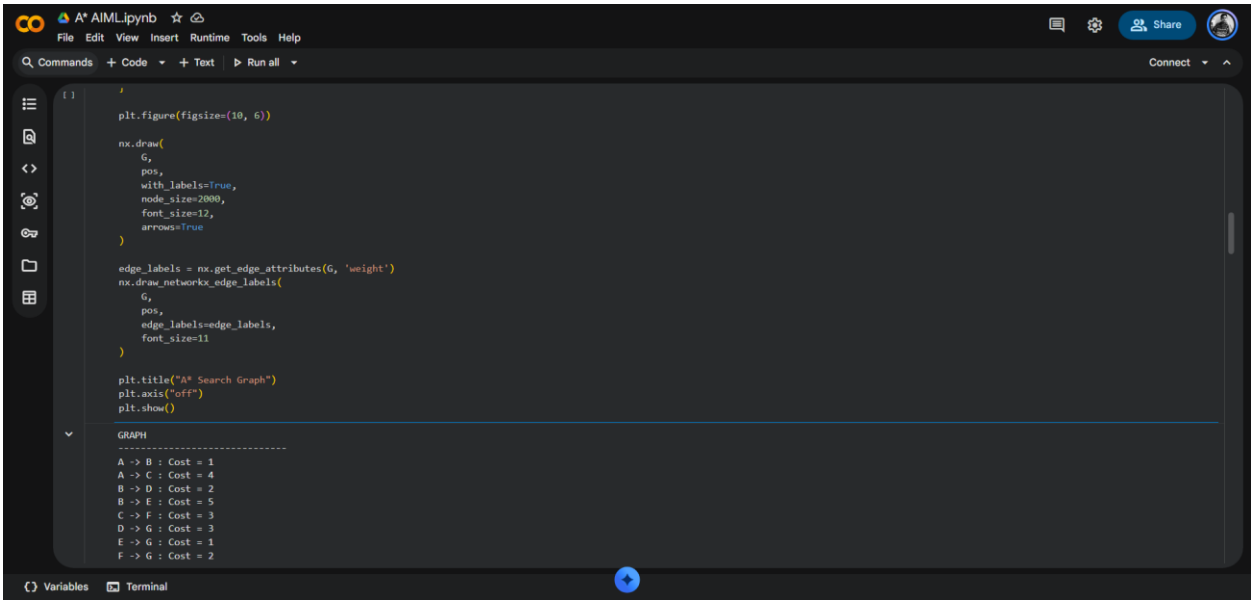

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!pip -q install networkx matplotlib
import heapq
import networkx as nx
import matplotlib.pyplot as plt
def a_star(graph, heuristic, start, goal):
open_list = [(heuristic[start], 0, start, [start])]
visited = set()
while open_list:
f, g, current, path = heapq.heappop(open_list)
if current in visited:
continue
visited.add(current)
if current == goal:
return path, g
for neighbor, cost in graph[current]:
if neighbor not in visited:
new_g = g + cost
new_f = new_g + heuristic[neighbor]
heapq.heappush(
open_list,
(new_f, new_g, neighbor, path + [neighbor])
)
return None, float('inf')
Variables
Terminal


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graph = {
'A': [('B', 1), ('C', 4)],
'B': [('D', 2), ('E', 5)],
'C': [('F', 3)],
'D': [('G', 3)],
'E': [('G', 1)],
'F': [('G', 2)],
'G': []
}
heuristic = {
'A': 6,
'B': 5,
'C': 4,
'D': 3,
'E': 1,
'F': 2,
'G': 0
}
start = 'A'
goal = 'G'
print("GRAPH")
print("-" * 30)
for node in graph:
for neighbor, cost in graph[node]:
print(f"{node} -> {neighbor} : Cost = {cost}")
print("\nHEURISTIC VALUES")
print("-" * 30)
for node, value in heuristic.items():
Variables
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print(f"{node} -> {neighbor} : Cost = {cost}")
print("\nHEURISTIC VALUES")
print("-" * 30)
for node, value in heuristic.items():
print(f"h({node}) = {value}")
path, cost = a_star(graph, heuristic, start, goal)
print("\nRESULT")
print("-" * 30)
print(f"Start Node : {start}")
print(f"Goal Node : {goal}")
print(f"Shortest Path : {' -> '.join(path)}")
print(f"Total Cost : {cost}")
G = nx.DiGraph()
for node in graph:
for neighbor, cost in graph[node]:
G.add_edge(node, neighbor, weight=cost)
pos = {
'A': (0, 2),
'B': (2, 3),
'C': (2, 1),
'D': (4, 3),
'E': (4, 1),
'F': (4, 0),
'G': (6, 2)
}
Variables
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plt.figure(figsize=(10, 6))
nx.draw(
G,
pos,
with_labels=True,
node_size=2000,
font_size=12,
arrows=True
)
edge_labels = nx.get_edge_attributes(G, 'weight')
nx.draw_networkx_edge_labels(
G,
pos,
edge_labels=edge_labels,
font_size=11
)
plt.title("A* Search Graph")
plt.axis("off")
plt.show()
GRAPH
A -> B : Cost = 1
A -> C : Cost = 4
B -> D : Cost = 2
B -> E : Cost = 5
C -> F : Cost = 3
D -> G : Cost = 3
E -> G : Cost = 1
F -> G : Cost = 2
Variables
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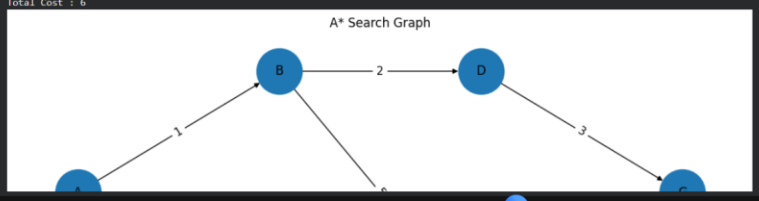
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```
plt.axis("off")
plt.show()
```

```
E -> G : Cost = 1
F -> G : Cost = 2

HEURISTIC VALUES
-----------------------------
h(A) = 6
h(B) = 5
h(C) = 4
h(D) = 3
h(E) = 1
h(F) = 2
h(G) = 0

RESULT
-----------------------------
Start Node : A
Goal Node  : G
Shortest Path : A -> B -> D -> G
Total Cost : 6
```



Variables Terminal

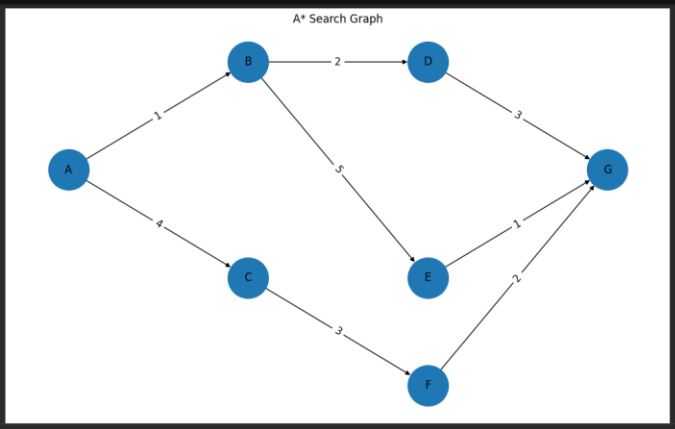
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```
!pip -q install networkx matplotlib
```

```
import heapq
import networkx as nx
```

Variables Terminal